The Cost of the Prompt: Evaluating SAM 2 for Prompt-Free Agricultural Field Segmentation

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1. INTRODUCTION

Segment Anything (SAM) [2] and SAM 2 [1] segment natural images well when a user clicks on the object. Remote-sensing pipelines have no user, so the prompt must be generated automatically, usually as a regular grid of points. We ask *what this prompt costs* when a frozen SAM 2.1 segments agricultural parcels in Sentinel-2 images [4], and whether adapting the mask decoder can pay that cost. We report three findings, two of them negative: a weak frozen baseline, a decoder fine-tune that collapses under automatic prompting, and a failure analysis that places the bottleneck in prompt density rather than in the decoder.

2. METHODOLOGY

Data. We use the Patches Assembled Dataset of Sen4AgriNet [3]: six Sentinel-2 tiles, three in Spain (ES) and three in France (FR), seeded random sampling, cloud-free May 2019 acquisitions [5]. This gives 888 RGB patches of 366 × 366 px (444 per country) with 211,375 parcels of at least 20 px (median 113 px, about 11 × 11 px). Split by patch, stratified by tile: 624 train / 132 validation / 132 test.

Model and metric. Frozen SAM 2.1 with a Hiera-Small encoder [6] and the automatic mask generator (16 × 16 point grid, predicted-IoU threshold 0.5, stability threshold 0.8). Each ground-truth parcel is scored by its best IoU over all generated masks; we report the mean over parcels (mean best IoU) and the fraction of parcels with IoU > 0.5.

Phase A: teardown. Forward hooks measure parameters and MACs. The image encoder holds 74.5 % of the parameters and the mask decoder 9.2 %. The encoder costs 133.6 GMACs once per image, the decoder 1.37 GMACs per prompt; at 16 × 16 points the decoder already costs 2.6× the encoder, at 32 × 32 10.5×. In prompt-free mode the prompt, not the backbone, dominates compute.

Phase B: decoder adaptation. We train only the second two-way transformer layer and the IoU head of the mask decoder (1.71 M parameters, 3.7 % of the model) with a Dice loss, encoder frozen. v1 trained on parcel-centroid prompts and selected the checkpoint by validation loss. v2 is a controlled ablation with expectations written before the results: training prompt (centroid, random point, generator grid point) × loss on token 0 or on tokens 1–3, selected by the generator's own validation IoU, plus a v1 replicate selected by loss.

Phase C: failure analysis. We overlay the generator grid on the true parcel masks and count the parcels that receive at least one point, by size and country; on the frozen model we sweep 16/32/64 points per side with runtime, and evaluate with both filters off.

3. RESULTS

Baseline. The frozen model reaches a mean best IoU of 0.161 on all 888 patches; only 14.6 % of parcels exceed IoU 0.5. Spain is much worse than France (field-weighted 0.085 vs. 0.220). Qualitatively, SAM finds field boundaries but often merges neighbouring parcels.

v1 collapses under automatic prompting. On validation with centroid prompts, mask IoU rose from 0.312 to 0.530. On the 132 test patches with the automatic generator it fell from 0.156 to 0.064 (paired

Δ = −0.092, 95 % bootstrap CI [−0.103, −0.083]; 128 of 132 patches worse), and masks per patch fell from 78.7 to 41.1. The training prompt was not the inference prompt, and a validation loss on the training prompt could not detect it.

The grid misses most parcels (Fig. 1). A 16 × 16 grid places a point every 22.9 px, so only 32 % of parcels receive any point (ES 24 %, FR 40 %; parcels of 20–49 px: 6 %). Per-patch IoU correlates with median parcel area (Spearman ρ = 0.87, confounded with country). No decoder can segment a parcel that receives no prompt: this is the cost of the prompt.

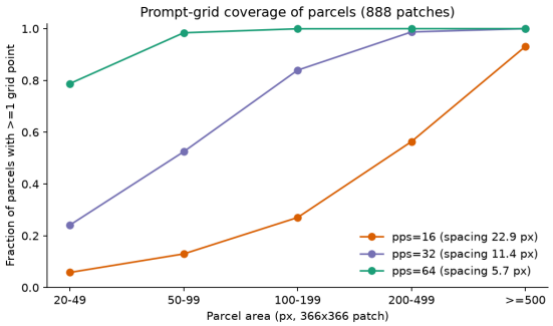


Fig. 1. Fraction of parcels with at least one generator grid point, by parcel area, for 16, 32 and 64 points per side (888 patches).

Paying the cost (Table 1). On the frozen model, denser grids give mean best IoU [PPS16_IOU] / [PPS32_IOU] / [PPS64_IOU] at 16 / 32 / 64 points per side. With both filters off it reaches [FILTERS_OFF_IOU]. The best v2 run ([BEST_RUN]) reaches [BEST_RUN_IOU] against 0.156 untrained; the loss-selected v1 replicate reaches [A_LOSS_SELECT_IOU].

Table 1. Grid density on the frozen model: point spacing on a 366 px patch, fraction of parcels with at least one grid point (all 888 patches), mean best IoU and seconds per patch (test subset).

Points per side	Spacing (px)	Parcels hit	Mean best IoU	s / patch
16	22.9	32 %	[PPS16_IOU]	[PPS16_TIME]
32	11.4	68 %	[PPS32_IOU]	[PPS32_TIME]
64	5.7	95 %	[PPS64_IOU]	[PPS64_TIME]

4. CONCLUSIONS

Prompt-free SAM 2.1 transfers poorly to Sentinel-2 parcels, and the dominant reason is the automatic prompt: at the default density two thirds of the parcels, and almost all small Spanish ones, never receive a point, while each extra point costs a full decoder pass. Adapting the decoder attacks the wrong bottleneck, and doing so with a prompt that differs from the inference prompt actively destroys the generator; model selection must use the generator itself. **Limitations:** one seed; RGB only; one month; 22 test patches per tile; grid coverage counts a point inside a parcel, not a correct mask; the ES/FR gap mixes parcel size with crop and radiometric shift.

REFERENCES

[1] N. Ravi et al., “SAM 2: Segment Anything in Images and Videos,” arXiv:2408.00714, 2024.
[2] A. Kirillov et al., “Segment Anything,” in Proc. IEEE/CVF ICCV, pp. 4015–4026, 2023.
[3] D. Sykas, M. Sdraka, D. Zografakis and I. Papoutsis, “A Sentinel-2 Multiyear, Multicountry Benchmark Dataset for Crop Classification and Segmentation With Deep Learning,” IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens., vol. 15, pp. 2937–2953, 2022.
[4] M. Drusch et al., “Sentinel-2: ESA's Optical High-Resolution Mission for GMES Operational Services,” Remote Sens. Environ., vol. 120, pp. 25–36, 2012.
[5] A. Hollstein, K. Segl, L. Guanter, M. Brell and M. Enesco, “Ready-to-Use Methods for the Detection of Clouds, Cirrus, Snow, Shadow, Water and Clear Sky Pixels in Sentinel-2 MSI Images,” Remote Sens., vol. 8, no. 8, p. 666, 2016.
[6] C. Ryali et al., “Hiera: A Hierarchical Vision Transformer without the Bells-and-Whistles,” in Proc. ICML, 2023.